Lecture 7 Tutorial — Answers

Q1. Signed Distance to a Hyperplane

For

$$H = \{x : \boldsymbol{w}^\top x + b = 0\}, \qquad \boldsymbol{w} \neq 0,$$

the signed distance from a point x to H is

$$\mathrm{dist}_\pm(x, H) = \frac{\boldsymbol{w}^\top x + b}{\|\boldsymbol{w}\|}.$$

1. $$\mathrm{dist}_\pm(x, H) = \frac{\boldsymbol{w}^\top x + b}{\|\boldsymbol{w}\|}$$
2. The sign indicates **which side of the hyperplane** the point lies on.

Q2. Scale Invariance and the Need for Normalization

Consider

$$\hat{y} = \mathrm{sign}(\boldsymbol{w}^\top x + b).$$

1. Under $(\boldsymbol{w}, b) \mapsto (c\boldsymbol{w}, cb)$ with $c > 0$,
   $$(c\boldsymbol{w})^\top x + cb = c(\boldsymbol{w}^\top x + b).$$
   Hence
   $$(c\boldsymbol{w})^\top x + cb = 0 \iff \boldsymbol{w}^\top x + b = 0,$$
   so the decision boundary does not change.
2. The quantity scales as
   $$y_i\big((c\boldsymbol{w})^\top x_i + cb\big) = c\, y_i(\boldsymbol{w}^\top x_i + b).$$
3. Without normalization, the margin objective is not meaningful because it can be made arbitrarily large by scaling $(\boldsymbol{w}, b)$ by any $c > 0$ while leaving the classifier unchanged.

Q3. Geometric Margin

For

$$\gamma_i = \frac{y_i(\boldsymbol{w}^\top x_i + b)}{\|\boldsymbol{w}\|},$$

the margin of the classifier on the dataset is

$$\gamma = \min_i \gamma_i.$$

1. $$\gamma = \min_i \frac{y_i(\boldsymbol{w}^\top x_i + b)}{\|\boldsymbol{w}\|}$$
2. If the "unnormalized margin" is defined as $y_i(\boldsymbol{w}^\top x_i + b)$, then under scaling $(\boldsymbol{w}, b) \mapsto (c\boldsymbol{w}, cb)$ it becomes
   $$c\, y_i(\boldsymbol{w}^\top x_i + b),$$
   so it can be made arbitrarily large without changing the boundary; this is the degeneracy.

Q4. Hard-Margin SVM (Primal Formulation)

- Decision variables: $\boldsymbol{w}$, b
- Objective:

$$\min_{\boldsymbol{w},b} \frac{1}{2}\|\boldsymbol{w}\|^2$$

- Constraints:

$$y_i(\boldsymbol{w}^\top x_i + b) \geq 1, \qquad \forall i$$

So the primal problem is

$$\min_{\boldsymbol{w},b} \frac{1}{2}\|\boldsymbol{w}\|^2 \quad \text{subject to } y_i(\boldsymbol{w}^\top x_i + b) \geq 1,\ \forall i.$$

Q5. Margin Width

The supporting hyperplanes are

$$\boldsymbol{w}^\top x + b = 1 \qquad \text{and} \qquad \boldsymbol{w}^\top x + b = -1.$$

1. Their distance is

$$\frac{2}{\|\boldsymbol{w}\|}.$$

2. Since the separation width is

$$\frac{2}{\|\boldsymbol{w}\|},$$

   minimizing $\|\boldsymbol{w}\|$ maximizes the distance between the two supporting hyperplanes.

Q6. Convexity Check

1. **T**. Each constraint is a linear inequality in $(\boldsymbol{w}, b)$, so it defines a half-space, and the intersection of half-spaces is convex.
2. **T**. The objective

$$\frac{1}{2}\|\boldsymbol{w}\|^2$$

   is a convex quadratic.
3. **T**. In a convex optimization problem, any local minimum is also a global minimum.

Q7. Lagrangian and Stationarity

Rewrite constraints as

$$1 - y_i(\boldsymbol{w}^\top x_i + b) \leq 0, \qquad \alpha_i \geq 0.$$

1. The Lagrangian is

$$L(\boldsymbol{w}, b, \alpha) = \frac{1}{2}\|\boldsymbol{w}\|^2 + \sum_{i=1}^{n} \alpha_i\big(1 - y_i(\boldsymbol{w}^\top x_i + b)\big).$$

2. Stationarity conditions:

$$\frac{\partial L}{\partial \boldsymbol{w}} = \boldsymbol{w} - \sum_{i=1}^{n} \alpha_i y_i x_i = 0$$

   so

$$\boldsymbol{w} = \sum_{i=1}^{n} \alpha_i y_i x_i.$$

Also,

$$\frac{\partial L}{\partial b} = -\sum_{i=1}^{n} \alpha_i y_i = 0$$

so

$$\sum_{i=1}^{n} \alpha_i y_i = 0.$$

3. Therefore,

$$\boldsymbol{w} = \sum_{i=1}^{n} \alpha_i y_i x_i, \qquad \sum_{i=1}^{n} \alpha_i y_i = 0.$$

Q8. Dual Problem

The hard-margin dual problem is

$$\max_{\alpha} \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j x_i^\top x_j$$

subject to

$$\alpha_i \geq 0, \qquad \sum_{i=1}^{n} \alpha_i y_i = 0.$$

1. Only the training points with $\alpha_i > 0$ can influence

$$\boldsymbol{w} = \sum_{i} \alpha_i y_i x_i.$$

2. If only a few α_i are nonzero, the representation is **sparse**; only a few points determine the classifier, and these are the support vectors.

Q9. KKT Complementary Slackness

Complementary slackness:

$$\alpha_i \big(1 - y_i(\boldsymbol{w}^\top x_i + b)\big) = 0.$$

1. If

$$y_i(\boldsymbol{w}^\top x_i + b) > 1,$$

then

$$\alpha_i = 0.$$

2. If

$$y_i(\boldsymbol{w}^\top x_i + b) = 1,$$

then the constraint is active, so typically

$$\alpha_i > 0.$$

Support vectors are exactly the points such that $\alpha_i > 0$.

Q10. Recovering from Support Vectors

Given

$$\boldsymbol{w} = (2, -1)^\top, \qquad x_k = (1, 2)^\top, \qquad y_k = +1,$$

use

$$b = y_k - \boldsymbol{w}^\top x_k.$$

Compute:

$$\boldsymbol{w}^\top x_k = (2, -1) \cdot (1, 2) = 2 - 2 = 0.$$

Hence

$$b = 1 - 0 = 1.$$

So,

$$\boxed{b = 1}$$

Q11. Soft-Margin SVM and Slack Interpretation

Soft-margin constraints:

$$y_i(\boldsymbol{w}^\top x_i + b) \geq 1 - \xi_i, \qquad \xi_i \geq 0.$$

Equivalently,

$$\xi_i = \max(0, 1 - y_i f(x_i)).$$

Given

$$y_1 f(x_1) = 1.2, \qquad y_2 f(x_2) = 0.7, \qquad y_3 f(x_3) = -0.3,$$

we obtain:

1.
$$\xi_1 = \max(0, 1 - 1.2) = 0$$
$$\xi_2 = \max(0, 1 - 0.7) = 0.3$$
$$\xi_3 = \max(0, 1 - (-0.3)) = 1.3$$

2. Classification:
   - $y_1 f(x_1) = 1.2$: **(A) outside/on margin (no violation)**
   - $y_2 f(x_2) = 0.7$: **(B) inside margin but correctly classified**
   - $y_3 f(x_3) = -0.3$: **(C) misclassified**

Q12. Kernel SVM Decision Function

Given

$$f(x) = \sum_{i \in S} \alpha_i y_i K(x_i, x) + b, \qquad \hat{y} = \operatorname{sign}(f(x)),$$

with linear kernel

$$K(x, z) = x^\top z,$$

and

$$x_1 = (1, 0),\ y_1 = +1,\ \alpha_1 = 2,$$
$$x_2 = (0, 1),\ y_2 = -1,\ \alpha_2 = 1,$$
$$b = -0.5, \qquad x = (2, 1),$$

compute:

$$K(x_1, x) = (1, 0) \cdot (2, 1) = 2,$$
$$K(x_2, x) = (0, 1) \cdot (2, 1) = 1.$$

Thus

$$f(x) = 2(+1)(2) + 1(-1)(1) - 0.5 = 4 - 1 - 0.5 = 2.5.$$

1.

$$f(x) = 2.5$$

2. Since

$$f(x) > 0,$$

the predicted label is

$$\hat{y} = +1.$$